

PROFESSIONAL SUMMARY

Machine Learning Engineer with 3+ years of professional experience, including 2 years in AI/ML and 1 year in data operations. Experienced in designing and deploying RAG-based chatbots and voice bots, vector search engines, and real-time computer vision systems using LangChain, LangGraph, Neo4j, FastAPI, and cloud platforms such as AWS and Kubernetes. Skilled in data preprocessing, model development, and scalable deployment, delivering production-ready AI solutions that drive efficiency and improve user experiences across bioinformatics, e-commerce, and Automotive domains.

EXPERIENCE

AI/ML Engineer [\[Value Health Solutions, Coimbatore\]](#)

Dec 2025 – Present

- **Graph RAG Pipeline Development:** Introduced and led adoption of Graph Retrieval-Augmented Generation (Graph RAG) for enterprise RAG applications, designing a scalable knowledge graph-based retrieval pipeline with custom components for AWS Bedrock, Textract, and domain-specific data loaders to enable context-aware, multi-hop reasoning.
- **AI-Powered Revenue Cycle Management (RCM) Application:** Developed an AI-powered Revenue Cycle Management (RCM) dashboard that automates patient intake processing using OCR and AI agents, extracting and validating clinical and billing information from patient intake forms. Built intelligent workflows to calculate medical billing and cost estimation using CPT, ICD, and diagnosis codes, improving billing accuracy and operational efficiency. Designed AI agents to identify and map doctor information directly from patient intake forms and clinical records. Integrated and collaborated with 10+ US healthcare payers to streamline claims processing, eligibility verification, and billing workflows across healthcare operations.

Machine Learning Engineer [\[Kadit Innovations, Tenkasi, India\]](#)

Mar 2025 – Nov 2025

- **Protein Vector Search :** Designed and implemented an AI-powered protein search engine using Neo4j vector databases to perform similarity-based retrieval, integrating metadata from multiple protein databases to map sequences and identify relevant matches, including detailed annotation of post-translational modifications (PTMs).
- **CCTV Attendance Monitoring System:** Built a real-time attendance and security monitoring solution using RTSP camera feeds with on-device motion detection. Integrated WebSocket triggers to capture events, applied YOLO for person detection and InsightFace facial embeddings for identity recognition, and logged verified entries into PostgreSQL for attendance tracking and auditing.
- Deployed the applications on AWS, leveraging services such as EC2, S3, EKS and Lambda to ensure scalable performance and reliability for processing large biological datasets in a cloud-native environment.

Associate Machine Learning Engineer [\[ISPG Technologies Pvt.Ltd, Kochin, Kerala\]](#)

Dec 2024 – Mar 2025*(Intern: 7 Months → Associate ML Engineer: 3 Months)*

- **RAG-Based Product User-Manual Chatbot (Development):** Created a RAG-based product manual chatbot API by crawling and scraping over 5000 websites using Selenium, stored embeddings in Mongo-VectorSearch, and structured the workflow using LangGraph with tools and agents for particular sections.
- **AI Chatbot Workflow with LangGraph Agents & AWS (Pre-Production):** Developing a product with a structured workflow for an AI chatbot using LangGraph tools and agents, enabling efficient, task-specific interactions by automating responses and decision-making processes, improving user experience and achieving accuracy levels of over 70% compared to manual flow approaches. The solution features a centralized chatbot with an admin dashboard for controlling and customizing functionalities.
- **RAG Technology Integration (Development):** Developed and integrated Database Retrieval-Augmented Generation (RAG) into the CRM database, enabling seamless translation of user NLP queries into Cypher queries (Graph DB) and converting Cypher responses into human-readable formats. Implemented restricted delete, create, and update operations. This integration enhanced client interaction and operational efficiency by 40%.
- **AI-Based E-Commerce Query System (Production):** Designed and developed two AI-based e-commerce query systems and an admin dashboard for cost and query analysis using MongoDB VectorSearch for automobile spare parts and jewelry retailers. Implemented AI-driven search systems that enhanced customer experience with fast, accurate queries and natural language processing by 200ms. Integrated advanced search and AI recommendations, improving user experience by 50% and driving sales.

Data & Operations Associate [MK Labels, Chennai]

May 2021 – Jul 2022

- Collected, organized, and maintained large datasets of financial transactions (credit, debit, accounts payable/receivable) using Excel and Pandas, ensuring data accuracy and integrity.
- Managed inventory Data of 500,000+ stock units/Day using MySQL, ensuring 99% on-time delivery and accurate stock records.
- Streamlined workflows between clients and production teams using MySQL, Cron jobs, and automated email alerts, increasing operational efficiency by 20%.

EDUCATION

Bharathiar University, Coimbatore

Aug 2022 - Apr 2024

Masters in Data Analytics, Cumulative CGPA: 7.91/10

Ramakrishna Mission Vivekananda College, Chennai

Jun 2018 - Jun 2021

Bachelors in Physics, Cumulative CGPA: 8.02/10

SKILLS

Libraries	Python (Numpy, Pandas, Matplotlib, Seaborn, Scikit-Learn, Tensorflow, Keras, NLTK, Streamlit), Fast-API, Flask, Langchain, Langgraph, Neo4j
Languages	Python (Intermediate), R (Fundamentals), SQL, Cypher (Neo4J)
Data Wrangling	Data Preparation, Data Extraction, Data Cleaning, Exploratory Data Analysis(EDA), Data Visualization, Feature Engineering, Feature selection, Model Building, IoT.
Machine Learning	Data Modeling, Clustering & Classification, Quantitative Analysis, Regression, Transformers, Predictive Modeling, Model Validation, Model deployment, CNN & RNN, LSTM, BERT.
Tools	Jupyter Notebook, Arduino IDE, MS Excel, Tableau, VS code, Postman, n8n.
Technologies	Fundamentals (Hive), Langchain, LLM, Large Language Models, IoT, Onnx, Docker, Kubernetes.
Cloud	AWS EC2, AWS IAM, AWS Beanstalk, AWS Bedrock, AWS RDS, AWS Sagemaker (Fundamentals), AWS APIGateway, AWS Lambda, AWS S3, AWS ECR, GCP

PROJECTS

HUMAN VS AI TEXT CLASSIFICATION, [NLP, LLM, BERT, Transformers, Deep Learning] [\[Link\]](#)

January 2024

- Implemented XLMBert, a variant of the BERT architecture, using the Keras NLP framework for natural language processing tasks, capable of handling more than 100,000 instances.
- Conducted fine-tuning of the model for Adaptation to Domain and Data using a custom classification dataset, employing techniques such as stopword removal and word embeddings for preprocessing.
- Achieved 99% testing accuracy in discerning human-generated text from machine-generated text, demonstrating the model's effectiveness.

MRI Image Enhancer, [Deep Learning, CNN, Image Processing] [\[Link\]](#)

December 2023

- Implemented TensorFlow's CNN framework to enhance low resolution medical MRI images using the FSRCNN model increased image clarity by 35% enabling extraction of crucial details for accurate diagnosis and treatment planning.
- Developed a Streamlit environment for interactive deployment, facilitating easy input of images and immediate visualization of enhanced outputs. This streamlined workflow led to a 14% improvement in real-time predictions.
- The FSRCNN model with PReLU activation performs better compared to bicubic and ReLU by 20% and 2%, respectively.

COURSES AND CERTIFICATIONS

- Neo4j Certified Professional, **Neo4j**
- AWS Certified AI Practitioner AIF-C01, **Udemy**
- AWS Certified Cloud Practitioner CLF-C02, **Udemy**
- Cypher Fundamentals, **Neo4J**
- Database Management System, **NPTEL**